

DEFICIENCIES & WHAT TESTS YOU NEED

The Missing Piece Behind Weight Loss, Energy, and Keeping It Off

Most people are deficient or imbalanced in at least one major area and never know it because the right labs are not always requested. This guide explains the body systems, deficiencies, and tests that matter when someone is trying to lose weight, improve energy, reduce cravings, and maintain results long term.

This guide is for education and preparation. It does not diagnose, treat, cure, or replace medical advice. Lab results should be reviewed with a licensed medical professional, especially if you are pregnant, taking medication, have kidney disease, liver disease, thyroid disease, diabetes, high blood pressure, anemia, or any chronic condition.

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1. Why Weight Loss Stalls

Weight loss is not only about eating less and moving more. Food and activity matter, but the body also needs the right internal environment to release stored energy, control cravings, recover, sleep, balance fluids, digest food, and maintain lean tissue. When the thyroid, liver, kidneys, gut, hormones, blood sugar, inflammation, or nutrient stores are stressed, results may slow down even when someone is trying hard.

The purpose of testing is to stop guessing. A lab panel can reveal patterns that may explain low energy, cravings, stubborn belly fat, poor recovery, hair shedding, cold intolerance, water retention, bloating, low mood, and plateaus.

The goal is not just weight loss. The goal is creating a body foundation that can keep the weight off.

Common reasons progress stalls:

- Low thyroid output or poor thyroid conversion
- Insulin resistance or poor blood sugar control
- Low vitamin D, magnesium, B12, iron, zinc, or protein intake
- Overloaded liver pathways or poor bile flow
- Kidney stress, fluid imbalance, or electrolyte imbalance
- Chronic inflammation
- Poor digestion and low nutrient absorption
- Sleep and cortisol dysregulation
- Hormone imbalance

2. The Thyroid: Your Metabolic Control Center

The thyroid is a small gland in the neck, but it has a major influence on metabolism. People often say the thyroid controls everything from the neck down because thyroid hormones send signals throughout the body that affect energy production, body temperature, digestion, heart rhythm, brain focus, muscle recovery, hair growth, skin texture, and how efficiently the body uses fuel.

When thyroid signaling is low or conversion is poor, the body may feel like it is running on low battery. This can make weight loss feel slow, even with nutrition changes.

Signs the thyroid may need attention:

- Fatigue or low motivation
- Cold hands and feet
- Dry skin, brittle nails, or hair shedding
- Constipation
- Brain fog or low mood
- Weight gain or difficulty losing weight
- Slow heart rate, low body temperature, or feeling sluggish
- Heavy cycles or cycle changes in some people

Tests to request:

Test	What it looks at	Why it matters
TSH	Brain signal to the thyroid	High or very low values may show the thyroid system needs closer review.
Free T4	Storage form of thyroid hormone	Shows what the thyroid is producing before conversion.
Free T3	Active thyroid hormone	Often more connected to energy and metabolic symptoms.
Reverse T3	Inactive thyroid pathway	Can rise with stress, low calories, illness, or inflammation.
TPO antibodies	Autoimmune thyroid marker	Screens for Hashimoto pattern.
Thyroglobulin antibodies	Additional autoimmune marker	Helpful when symptoms exist but TPO is normal.

Thyroid support basics: adequate protein, selenium-containing foods such as Brazil nuts, iodine from food sources when appropriate, zinc, iron/ferritin support, vitamin D, stress regulation, enough calories, and sleep. Do not start iodine or thyroid supplements without medical guidance if you have thyroid disease or antibodies.

3. The Liver: Your Processing Plant

The liver is one of the most important organs for metabolism and transformation. It processes nutrients, stores vitamins, helps regulate blood sugar, produces bile for fat digestion, processes hormones, makes cholesterol needed for hormone production, and supports detoxification pathways.

The liver does not just “detox.” It helps decide how nutrients are used, stored, converted, and cleared. If liver enzymes are elevated, bile flow is sluggish, or the liver is overwhelmed by alcohol, excess sugar, fatty liver patterns, medication stress, or inflammation, weight management can become harder.

Signs liver support may be needed:

- Bloating after fatty meals
- Nausea or heaviness after eating
- Skin issues or itching
- Brain fog
- Stubborn belly weight
- Poor recovery
- High triglycerides or cholesterol changes
- Right side discomfort under ribs - seek care if severe

Liver labs to request:

Test	What it looks at	Why it matters
ALT	Liver cell stress	Can rise with fatty liver, medication stress, alcohol, or inflammation.
AST	Liver/muscle enzyme	Interpreted with ALT and symptoms.
GGT	Bile/liver stress marker	Can be useful when fatty liver, alcohol, or bile stress is suspected.

Test	What it looks at	Why it matters
Alkaline phosphatase	Bile flow/bone marker	Can suggest bile flow patterns when paired with other labs.
Bilirubin	Breakdown/clearance marker	Helps evaluate bile and liver processing.
Lipid panel	Cholesterol/triglycerides	High triglycerides may connect to blood sugar and liver fat patterns.

Food and herb support: cruciferous vegetables, beets, leafy greens, citrus, garlic, onions, choline-rich foods, dandelion root, milk thistle, burdock root, turmeric, ginger, and adequate hydration. Herbs can interact with medications, pregnancy, gallbladder disease, and liver conditions, so use professional guidance when needed.

4. The Kidneys: Your Filtration System

The kidneys filter blood, regulate water balance, manage electrolytes, support blood pressure, help maintain pH balance, activate vitamin D, and assist red blood cell signaling. For weight loss, kidney health matters because hydration, minerals, sodium/potassium balance, blood pressure, muscle function, energy, and water retention are all connected to kidney function.

When the kidneys are stressed, the body may hold fluid, blood pressure may rise, electrolytes may shift, and fatigue or muscle cramps may show up. This does not mean every symptom is kidney disease, but kidney labs should be part of a complete transformation panel.

Signs kidney or fluid balance may need review:

- Puffy eyes, swollen feet, or water retention
- Frequent urination or changes in urine appearance
- High blood pressure
- Muscle cramps or weakness
- Unusual fatigue
- Foamy urine or dark urine
- Back/flank pain or painful urination - seek medical care

Kidney labs to request:

Test	What it looks at	Why it matters
Creatinine	Kidney filtration marker	Used to estimate kidney filtration.
eGFR	Estimated kidney filtration rate	Important overview of kidney function.
BUN	Protein/waste processing marker	Can shift with hydration, protein intake, or kidney function.
Electrolytes	Sodium, potassium, chloride, CO2	Helps evaluate mineral and hydration balance.
Urinalysis	Urine screen	Checks protein, blood, glucose, ketones, infection signs.
Microalbumin/ACR	Small protein leakage	Early marker often used for kidney stress, especially with blood sugar or blood pressure issues.

Kidney support basics: consistent hydration, balanced minerals, moderate sodium, blood pressure management, blood sugar management, enough but not excessive protein for your health status, and medical review before using strong diuretics or herbs if kidney disease is possible.

5. Blood Sugar & Insulin Resistance

Insulin is the hormone that helps move glucose from the bloodstream into cells. When cells become less responsive to insulin, the body may produce more insulin to keep blood sugar controlled. High insulin patterns can make cravings, hunger, belly fat, and weight-loss resistance more common.

Signs of blood sugar imbalance:

- Sugar cravings
- Feeling hungry soon after eating
- Energy crash after meals
- Belly weight
- Waking at night hungry or wired
- Brain fog when meals are delayed
- Dark skin patches around neck or underarms in some people

Blood sugar labs:

Test	What it looks at	Why it matters
Fasting glucose	Blood sugar at one point in time	Useful baseline but may miss early insulin resistance.
Fasting insulin	How much insulin is needed fasting	Can reveal early metabolic stress before glucose rises.
Hemoglobin A1C	Average blood sugar over about 3 months	Helps track long-term glucose patterns.
C-peptide	Insulin production marker	Sometimes used when insulin production needs review.
Lipid panel	Triglycerides/HDL pattern	Can reflect insulin resistance patterns.

Support basics: protein at meals, fiber, walking after meals, strength training, sleep, hydration, magnesium, chromium from food, cinnamon, berberine only with medical guidance, and reducing constant grazing.

6. Gut Health & Nutrient Absorption

You can eat healthy food and still struggle if digestion and absorption are weak. The gut breaks food down, absorbs nutrients, houses immune tissue, supports hormone clearance through bowel movements, and communicates with the brain through the gut-brain axis.

Signs gut health may be affecting results:

- Bloating, gas, constipation, diarrhea
- Food sensitivity patterns
- Reflux or burping after meals
- Low appetite or constant cravings
- Skin flare-ups
- Fatigue after eating
- Low iron, B12, or magnesium despite supplementation

Tests to consider:

Test	What it looks at	Why it matters
Comprehensive stool test	Digestion, inflammation, microbes	Useful for chronic bloating or bowel changes.
H. pylori test	Bacteria linked to stomach symptoms	May affect digestion and nutrient absorption.
Celiac screen	Gluten-related autoimmune marker	Important with chronic digestive issues, anemia, or family history.
Food sensitivity testing	Possible reactive foods	Not always definitive, but may guide short-term elimination plans.

Support basics: fiber, fermented foods if tolerated, cooked vegetables, digestive rhythm, hydration, chewing slowly, regular bowel movements, magnesium support, and not overloading supplements before digestion is stable.

7. Inflammation & Water Retention

Inflammation is a normal healing response, but chronic inflammation can affect energy, recovery, water retention, joint comfort, insulin signaling, and overall wellness. Inflammation may come from stress, poor sleep, injuries, infections, gut issues, food reactions, blood sugar problems, excess alcohol, and environmental exposures.

Signs inflammation may be high:

- Joint discomfort or stiffness
- Puffiness and water retention
- Brain fog
- Fatigue
- Slow recovery after workouts
- Skin flare-ups
- Elevated blood pressure or glucose patterns

Inflammation labs:

Test	What it looks at	Why it matters
hs-CRP	Systemic inflammation	Sensitive marker used in metabolic and cardiovascular risk review.
ESR	General inflammation	Less specific but helpful with symptoms.
Ferritin	Iron storage and inflammation pattern	Can be low from deficiency or high with inflammation.
Omega-3 index	Fatty acid balance	May help assess anti-inflammatory fat status.

Support basics: omega-3 rich foods, colorful produce, turmeric, ginger, sleep, walking, strength training, stress regulation, oral health, gut support, and identifying trigger foods or exposures.

8. Hormones & Body Composition

Hormones act as messengers that affect hunger, fullness, cravings, sleep, mood, libido, muscle, recovery, fat distribution, and fluid balance. Hormone testing should be interpreted based on sex, age, symptoms, cycle timing when applicable, medications, and medical history.

Hormones to review:

Hormone/Test	Why it matters	Possible pattern if imbalanced
Cortisol	Stress rhythm and blood sugar support	Belly weight, poor sleep, cravings, fatigue.
DHEA-S	Adrenal androgen reserve	Low energy or low resilience patterns.
Testosterone	Muscle, libido, motivation, metabolism	Low muscle tone or high androgen symptoms depending on level.
Estradiol	Mood, fluid, cycle, skin, bone support	High/low patterns can affect weight and water retention.
Progesterone	Cycle rhythm, sleep, calm support	Low patterns may affect sleep, PMS, heavy cycles.
SHBG	Hormone binding protein	Can shift with insulin resistance, thyroid, liver status.

For men, low testosterone patterns may affect muscle, energy, mood, and fat distribution. For women, estrogen/progesterone/androgen patterns may affect cravings, cycle symptoms, water retention, and body composition. For everyone, cortisol and insulin patterns matter.

9. Common Deficiencies That Affect Results

Deficiencies can make it harder to feel motivated, build muscle, sleep well, manage cravings, and recover. Deficiencies do not always cause weight gain by themselves, but they can weaken the foundation needed for lasting results.

Deficiency	Common signs	Test to request
Vitamin D	Low mood, fatigue, immune issues, muscle aches	Vitamin D, 25-Hydroxy
Magnesium	Cramps, poor sleep, constipation, cravings, anxiety	RBC Magnesium
Vitamin B12	Brain fog, fatigue, tingling, low motivation	B12 plus methylmalonic acid if needed
Iron/Ferritin	Hair shedding, dizziness, cold, fatigue, poor stamina	Ferritin, iron panel, CBC
Zinc	Low immunity, hair/skin issues, poor taste/smell, cravings	Plasma zinc with copper review
Omega-3 fats	Inflammation, dry skin, mood issues, joint discomfort	Omega-3 index
Protein	Low muscle, cravings, slow recovery, hair changes	Diet review, albumin/prealbumin if clinically needed
Potassium/electrolytes	Cramps, weakness, fluid imbalance	CMP/electrolyte panel

Important: more is not always better. Iron, iodine, potassium, vitamin A, and certain herbs can be harmful when taken incorrectly or with certain health conditions. Testing helps personalize support.

10. What Tests To Request

Use this as a practical conversation list for your healthcare provider. You may not need every test at once, but this is a complete transformation-focused panel.

Category	Labs to request
Basic foundation	CBC, CMP, lipid panel, urinalysis

Category	Labs to request
Thyroid	TSH, Free T3, Free T4, Reverse T3, TPO antibodies, Thyroglobulin antibodies
Blood sugar/metabolism	Fasting glucose, fasting insulin, Hemoglobin A1C, C-peptide if needed
Liver/gallbladder pattern	ALT, AST, GGT, bilirubin, alkaline phosphatase, lipid panel
Kidney/fluid balance	Creatinine, eGFR, BUN, electrolytes, urinalysis, microalbumin/ACR
Nutrients	Vitamin D, B12, folate, ferritin, iron panel, RBC magnesium, zinc, omega-3 index
Inflammation	hs-CRP, ESR, ferritin review
Hormones	Cortisol AM or 4-point saliva, DHEA-S, testosterone, estradiol, progesterone, SHBG
Gut when needed	Stool analysis, H. pylori test, celiac screen, food sensitivity review

11. What Your Results May Mean

Lab interpretation is about patterns, not one number alone. A result can be technically “normal” but still not optimal for how someone feels. Also, symptoms matter. Review your results with a qualified provider who can interpret them based on your history, medications, and goals.

Pattern	What it may suggest	Possible next step
High fasting insulin with normal glucose	Early insulin resistance pattern	Protein/fiber meals, movement after meals, provider review.
Low ferritin	Low iron stores	Find cause before supplementing; review cycles, gut, diet.
Low vitamin D	Low vitamin D stores	Sunlight, food, supplement plan with follow-up testing.
High ALT/GGT	Liver stress or fatty liver pattern	Provider review, alcohol/sugar review, metabolic panel follow-up.
Low eGFR or protein in urine	Kidney function needs medical review	Do not self-treat; consult provider promptly.
High TSH with low Free T3/T4	Possible low thyroid output pattern	Thyroid provider review.
High hs-CRP	Inflammation pattern	Look for infection, gum health, gut, metabolic issues, injury, stress.

12. Food, Herbs & Lifestyle Support

Support should match the system that needs help. Food, hydration, sleep, and movement are the foundation. Herbs and supplements can be useful, but they should not be used blindly or as a replacement for testing.

System	Foods that support it	Herbs/support options
Thyroid	Protein, seafood, eggs, Brazil nuts, pumpkin seeds, iron-rich foods	Ashwagandha may not fit everyone; iodine only with guidance.
Liver	Cruciferous vegetables, beets, citrus, garlic, onions, greens	Milk thistle, dandelion root, burdock, turmeric, ginger.
Kidneys	Water, berries, cabbage, cauliflower, parsley, balanced minerals	Avoid strong diuretic herbs if kidney disease, pregnancy, or medication use.

System	Foods that support it	Herbs/support options
Blood sugar	Protein, fiber, beans, greens, berries, cinnamon foods	Cinnamon, berberine only with medical guidance.
Gut	Cooked vegetables, fiber, fermented foods if tolerated	Ginger, peppermint, chamomile, digestive bitters if appropriate.
Inflammation	Omega-3 fish, walnuts, berries, olive oil, colorful produce	Turmeric, ginger, green tea.

Daily lifestyle anchors:

- Eat protein with each meal to protect muscle and reduce cravings.
- Walk 10-15 minutes after meals when possible.
- Strength train 2-4 times weekly based on ability.
- Sleep 7-9 hours when possible and keep a consistent schedule.
- Hydrate and replace minerals appropriately, especially with sweating.
- Track bowel movements, energy, cravings, water retention, and sleep.
- Retest key labs after a consistent plan, often around 8-12 weeks, based on provider guidance.

13. Where To Get Testing

There are several ways to get labs. The best option depends on insurance, budget, location, and whether you want a clinician to interpret results.

Option	How it works	Best for
Primary care provider	Ask for labs during a visit.	Insurance coverage and medical interpretation.
Functional/integrative provider	Often orders deeper panels.	Detailed wellness and root-cause review.
Direct-to-consumer labs	Order some labs yourself through services connected to local draw sites.	Paying cash and getting quick access.
Quest/Labcorp draw sites	Used by many providers and self-order platforms.	Standard blood and urine labs.
At-home kits	Finger prick, saliva, urine, or stool kits mailed in.	Convenience for hormones, stool, vitamin D, A1C, or some wellness panels.

Examples of places people commonly look for at-home or self-pay testing include Everlywell, LetsGetChecked, Labcorp OnDemand, Quest, Walk-In Lab, and local pharmacy-based options. Availability, test menus, pricing, and state rules can change, so confirm directly before purchasing.

For urgent symptoms such as chest pain, severe weakness, fainting, severe abdominal pain, blood in urine, confusion, very high blood pressure, or signs of severe dehydration, do not use at-home testing as a substitute for medical care.

14. 30-Day Transformation Action Plan

This plan helps organize the first month before and after testing.

- 1 **Week 1 - Track your baseline:** energy, sleep, cravings, bowel movements, water retention, mood, workouts, weight, waist measurement, and symptoms.
- 2 **Week 1 - Request labs:** use the checklist and decide which tests you can access now.
- 3 **Week 2 - Build the plate:** protein, fiber, color, hydration, and minerals daily.
- 4 **Week 2 - Movement:** walk after meals and begin strength training at your level.
- 5 **Week 3 - Review patterns:** compare symptoms with labs. Look for thyroid, liver, kidney, insulin, inflammation, gut, and deficiency clues.
- 6 **Week 3 - Target support:** choose support based on findings, not guessing.
- 7 **Week 4 - Maintain:** repeat habits, adjust food, improve sleep, and plan retesting with your provider.

15. Final Lab Checklist

Category	Tests
Basic	CBC, CMP, lipid panel, urinalysis
Thyroid	TSH, Free T3, Free T4, Reverse T3, TPO antibodies, Thyroglobulin antibodies
Liver	ALT, AST, GGT, bilirubin, alkaline phosphatase
Kidneys	Creatinine, eGFR, BUN, electrolytes, microalbumin/ACR
Blood sugar	Fasting glucose, fasting insulin, A1C
Nutrients	Vitamin D, B12, folate, ferritin, iron panel, RBC magnesium, zinc, omega-3 index
Inflammation	hs-CRP, ESR
Hormones	Cortisol, DHEA-S, testosterone, estradiol, progesterone, SHBG
Gut	Stool analysis, H. pylori, celiac screen, food sensitivity review when appropriate

Final note: Your results are a map. Once you know what is low, high, stressed, or out of rhythm, you can stop guessing and build a plan that supports lasting change.